

Ma. Angeline T. Tipa

BS Computer Engineering (Big Data) | Polytechnic University of the Philippines - Manila

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Computer Engineering graduate with QA internship experience, targeting data analyst and data engineering roles.

EXPERIENCE

Quality Assurance On-the-Job Trainee

Jul 2025 – Sep 2025

DOST-CO-PES, Information Technology Division (Project LODI), Taguig City

- Wrote and executed 20+ test cases per screen across 17 UI screens and two user roles in two government systems, DIMIT and ISSP.
- Traced input validation defects to Screen Design Specifications, covering Ñ/ñ handling, blank inputs, URL format rules.
- Logged test cases in structured Excel with expected and actual results, pass or fail status, tester, date, and remarks.
- Collaborated with a senior technical specialist in review sessions and applied feedback to improve test coverage.

IT On-the-Job Trainee

Jul 2024 – Sep 2024

Local Government of Cavite, Management Information Service Department, Cavite City

- Built a basic Django and MySQL CRUD app with validated dropdowns for gender, suffix, and barangay instead of free text.
- Practiced basic SQL queries and Excel data encoding, gaining hands-on experience with structured data and database fundamentals.
- Presented Big Data and Machine Learning fundamentals to OJT supervisors and a fellow trainee as an assigned learning exercise.

Special Program for Employment of Students (SPES) — Beneficiary Records

Oct 2022 – Dec 2022

City Social Welfare & Development Office (CSWDO), Cavite City

- Encoded beneficiary records from hardcopy forms into Excel masterlists across barangays for a government assistance program.
- Validated ID eligibility and reconciled names against the masterlist, flagging expired IDs and duplicate entries.
- Facilitated assistance and scholarship payouts by distributing claimant forms, directing beneficiaries, and organizing lines.

SELECTED PROJECTS

Aralite - Analysis & dashboard on Philippine school enrollment | *Rebuilt a Big Data course activity* | aralite.vercel.app

- Analyzed DepEd's SY 2023 to 2024 enrollment data covering 60,167 schools and 27 million learners and found that the average senior high school offers only 2.52 of 8 tracks and strands.
- Found that learners in BARMM are over 6 times more likely to attend a single track or strand school than those in Region IV A.
- Built a Python and pandas ETL pipeline converting Excel to Parquet, then reshaping 58 columns into rows with zero dropped records.
- Validated published figures using notebook assertions, 32 unit tests, and an independent Excel reconciliation.
- Built a React and DuckDB WASM dashboard that runs SQL in the browser and lets users filter enrollment data by region.

Presyo - Automated daily data pipeline | *Personal project* | presyo-today.vercel.app

- Built a scheduled ETL pipeline with GitHub Actions pulling live rates into Supabase Postgres, tracking 8 currencies since August 2026.
- Added validation at the row level that rejects stale feeds, future dates, and out of band values, logging the reason to a public run log.

BIO-FISH - IoT control app for a bioplastic machine | *Capstone project, team of 4* | biofish-control.vercel.app

- Led a React Native (Expo) app controlling an ESP32 machine that turns fish scale waste into bioplastic across four automated stages.
- Served as Project Manager for the capstone's first semester phase, tracking a 10-week Gantt chart to 94% completion across fabrication, ethics compliance, and machine development planning.
- Modeled machine commands as a Supabase Postgres queue instead of a status flag, keeping commands reliable despite WiFi drops.

COURSEWORKS

MySQL Replication System | *Database Management Course*

- Designed and implemented Master–Master and Master–Slave replication with real-time multi-node data sync

Enrollment Data Dashboard | *Big Data Course* | *precursor to Aralite*

- Cleaned and validated a raw enrollment dataset, first manually then through a Python pipeline I helped build and catch errors.
- Built an automated Flask upload and cleaning workflow, letting the team add new yearly datasets without manual rework.
- Helped refine the Pandas and Dash dashboard interface for non technical viewers.

EDUCATION

Polytechnic University of the Philippines (PUP), Manila

2022 – 2026

- BS Computer Engineering (Big Data) | GWA: 1.50 | Media Head - Software Engineering Day 2026 (ASCEND).

TECHNICAL SKILLS

Data and Analytics: Python, pandas, NumPy, Matplotlib, Seaborn, SQL, PostgreSQL, MySQL, DuckDB, Power BI, Excel

Data Engineering: ETL Pipelines, Parquet, GitHub Actions Scheduling, Data Validation & Quality Reporting, MySQL replication

Software and Testing: React, TypeScript, React Native, Supabase, Git and GitHub, Vitest, Vercel, Manual QA and Test Case Design

CERTIFICATIONS

- **DataCamp (2026):** Python Data Fundamentals (7 course skill track), GitHub Foundations (4 course skill track), Design in Power BI (2 course skill track), Data Literacy Professional (8 course skill track), Intermediate SQL, Introduction to Excel
- **Cisco Networking Academy (2025):** CCNA - Introduction to Networks

AWARDS & HONORS

Academic & scholarship

- BIO-FISH, Top 8 finalist, 4th Annual Prototyping Exhibit, Awards & Recognition (APEAR 2026)
- DOST-JLSS Scholar (2024), qualified via a competitive scholarship examination
- 1st Place, Science Investigatory Project (school-level, 2015)
- 1st Place, Feature Writing, Division Schools Press Conference; RSPC Regional delegate (2015)

Competitions & activities

- YES-O Member (2016-2019)
- Bronze Medalist, Arnis Combative (STCAA Regional, (2017)
- Table Tennis competitor (division-level, 2016)
- Silver Medalist, Arnis Solo Baston (division-level, 2015)
- Arnis Yellow Belt Promotion (2015)
- 1st Place, Poster Making, Bulilit & Teen Health Workers (division-level, 2015)
- 1st Place, Kulay at Tubig Regada Painting Competition (division-level, 2015)
- Campus Journalism Workshop (school-level, 2015), multiple top placements